

Chapter 1

Introduction

1.1 Overview

The pharmaceutical industry plays a vital role in ensuring public health by providing safe and effective medicines. The supply chain involved in delivering these medicines is complex, consisting of multiple stakeholders such as manufacturers, distributors, wholesalers, and pharmacies. Due to the involvement of many intermediaries, maintaining transparency, security, and proper tracking of medicines becomes a challenging task.

Traditional pharmaceutical supply chain systems are mostly centralized and lack real-time tracking capabilities. This often leads to issues such as data manipulation, lack of transparency, and the circulation of counterfeit drugs. Counterfeit medicines pose serious risks to patients and reduce trust in healthcare systems. Additionally, centralized systems are vulnerable to single points of failure and unauthorized access.

To address these challenges, this project proposes a blockchain-based paracetamol supply chain management system. Blockchain technology provides a decentralized and secure platform where all transactions are recorded in an immutable manner. Once data is stored on the blockchain, it cannot be altered, ensuring integrity and reliability.

In this system, each medicine batch is assigned a unique identifier, and important details such as product name, quantity, manufacturing date, expiry date, and ownership information are recorded on the blockchain. As the product moves through different stages, each transaction is updated, creating a complete traceable history. This allows stakeholders to verify the authenticity and origin of medicines easily.

The system also uses smart contracts to automate operations like batch creation and ownership transfer. These self-executing programs reduce manual effort, minimize errors, and improve efficiency. A user-friendly frontend interface is integrated to allow users to interact with

Overall, the proposed system enhances transparency, security, and traceability in the pharmaceutical supply chain, making it more reliable and efficient compared to traditional systems.

1.2 Motivation

The motivation behind this project stems from the growing concerns over the presence of counterfeit drugs and lack of transparency in the pharmaceutical supply chain. Counterfeit medicines have become a serious global issue, as they not only result in financial losses for companies but also pose significant health risks to patients by delivering ineffective or harmful treatments.

In traditional supply chain systems, data is usually stored in centralized databases, which are highly vulnerable to manipulation, unauthorized access, and system failures. Since multiple stakeholders are involved, there is often a lack of coordination and limited visibility across different stages of the supply chain. This makes it difficult to trace the origin of medicines or verify their authenticity, allowing fake products to enter the market more easily.

Another major concern is the absence of a reliable mechanism to track the movement of pharmaceutical products in real time. Without proper traceability, identifying errors, delays, or fraudulent activities becomes challenging. This reduces trust among stakeholders and affects the overall efficiency of the system.

This project is therefore motivated by the need to:

- Ensure the authenticity and safety of medicines
- Provide complete end-to-end traceability of pharmaceutical products
- Improve transparency and trust among all stakeholders
- Minimize the risks of fraud, data tampering, and counterfeit drugs

Blockchain technology provides an effective solution to these challenges by offering a decentralized, secure, and tamper-proof system. By using blockchain, all transactions can be recorded transparently and cannot be altered once validated, making it highly suitable for building a trustworthy pharmaceutical supply chain system.

1.3 Objective

The primary objective of this project is to develop a secure, transparent, and efficient pharmaceutical supply chain system using blockchain technology. The system aims to improve the reliability of medicine tracking and ensure that all transactions are recorded in a tamper-proof and trustworthy manner.

The specific objectives of the project are as follows:

- To design and implement a system that enables tracking of medicines at every stage of the supply chain, from manufacturing to final delivery
- To utilize blockchain technology for maintaining a decentralized and immutable record of all transactions
- To implement smart contracts that automate key operations such as batch creation, ownership transfer, and validation of data
- To ensure data integrity, security, and transparency, preventing unauthorized modifications and reducing the chances of fraud
- To develop a user-friendly interface that allows users to easily interact with the system and monitor the movement of pharmaceutical products
- To reduce the risk of counterfeit drugs entering the supply chain by enabling easy verification of product authenticity
- To improve accountability among stakeholders by maintaining a clear and traceable history of all transactions.

In addition to these, the project also aims to demonstrate how modern technologies like blockchain can be effectively applied in real-world scenarios to solve critical issues in the healthcare sector. By ensuring that every transaction is verifiable and permanently recorded, the system creates a trustworthy environment for all participants involved in the pharmaceutical supply chain.

1.4 Scope

The scope of this project focuses on the design and development of a blockchain-based pharmaceutical supply chain management system that enables secure tracking and management of medicines throughout their lifecycle. The system is intended to demonstrate how blockchain technology can be applied to improve transparency, traceability, and data security in the pharmaceutical domain.

The project primarily covers the following functionalities:

- Creation of medicine batches by manufacturers, where each batch is assigned a unique identifier
- Recording essential product details such as medicine name, quantity, manufacturing date, and expiry date on the blockchain
- Enabling tracking of ownership transfer as the product moves between different stakeholders such as distributors and retailers
- Maintaining a complete and immutable history of each transaction for every medicine batch
- Providing users with real-time visibility into the current status and movement of products
- Integration of a user-friendly frontend interface that allows interaction with the blockchain system using tools like MetaMask

The system is developed as a prototype application, mainly for academic and demonstration purposes. It does not include full-scale deployment features such as large user handling, advanced security layers for enterprise use, or integration with government regulatory systems.

Additionally, the project is limited to core supply chain functionalities and does not cover aspects like logistics optimization, inventory forecasting, or detailed financial transactions. However, it establishes a strong foundation for future expansion.

Future enhancements of the system may include:

- Integration with IoT devices for real-time environmental monitoring (e.g., temperature control of medicines)
- Use of QR codes or RFID tags for easy product identification and verification
- Development of a mobile application for wider accessibility
- Implementation of advanced analytics and reporting tools for better decision-making

Overall, the scope of this project is to provide a practical and scalable approach for improving pharmaceutical supply chain management using blockchain technology, while also leaving room for further advancements and real-world applications.

1.5 Existing System

The existing pharmaceutical supply chain systems are largely centralized and semi-manual, relying on traditional database management and paper-based documentation. These systems involve multiple independent stakeholders such as manufacturers, distributors, wholesalers, and retailers, each maintaining their own records. Due to the lack of a unified and integrated platform, data sharing between these entities is often inefficient and inconsistent.

One of the major drawbacks of the existing system is the lack of transparency across different stages of the supply chain. Since information is stored in isolated systems, stakeholders have limited visibility into the movement and status of pharmaceutical products. This makes it difficult to monitor the complete lifecycle of a medicine.

Another critical issue is the high risk of data manipulation and fraud. Centralized databases can be altered or accessed by unauthorized users, leading to potential tampering of important information such as product origin, batch details, or expiry dates. This creates opportunities for counterfeit medicines to enter the market.

The system also suffers from poor traceability, as tracking a product's journey from manufacturer to consumer is often time-consuming and unreliable. In case of defects or safety concerns, identifying the source of the problem becomes difficult, delaying corrective actions.

Additionally, the dependence on paper-based records or disconnected digital systems increases the chances of human error, data loss, and duplication. The presence of a single point of failure in centralized systems further makes them vulnerable to system crashes or cyberattacks.

Due to these limitations, the current system struggles to ensure the authenticity, safety, and reliability of pharmaceutical products, ultimately affecting patient trust and healthcare quality.

1.6 Proposed System

The proposed system is a blockchain-based pharmaceutical supply chain management system designed to overcome the limitations of traditional systems by providing enhanced security, transparency, and traceability. The system leverages blockchain technology to create a decentralized platform where all transactions related to pharmaceutical products are securely recorded and cannot be altered once validated.

In this system, each medicine batch is created by the manufacturer and assigned a unique batch ID, along with important details such as product name, quantity, manufacturing date, and expiry date. These details are stored on the blockchain, ensuring that the data remains tamper-proof and permanently accessible. As the product moves through different stages of the supply chain—such as distributors and retailers—each transfer of ownership is recorded as a new transaction on the blockchain.

A key component of the proposed system is the use of smart contracts, which automate various processes such as batch creation, validation, and ownership transfer. These self-executing contracts ensure that predefined conditions are met before any transaction is completed, reducing manual intervention and minimizing errors. This automation improves efficiency and ensures consistency across the system.

The system provides real-time tracking and complete traceability of pharmaceutical products. Stakeholders can view the entire history of a medicine batch, from its origin to its current location, enabling easy verification of authenticity. This significantly reduces the risk of counterfeit drugs entering the supply chain.

Additionally, the system follows a decentralized architecture, eliminating the dependency on a central authority and reducing the risk of system failure or data breaches. A user-friendly frontend interface is integrated to allow users to interact with the blockchain network easily. Tools such as Ethereum, Hardhat, and MetaMask are used to facilitate blockchain operations and secure user interactions.

Overall, the proposed system enhances trust, accountability, and efficiency among all stakeholders by ensuring that every transaction is transparent, verifiable, and permanently recorded. It provides a reliable solution for improving the safety and integrity of the pharmaceutical supply chain

Chapter 2

Problem Statement

2.1 Problem Statement

The pharmaceutical supply chain is a complex network involving multiple stakeholders such as manufacturers, distributors, wholesalers, and retailers. Each of these entities operates independently, often maintaining separate records and systems. In the current scenario, there is no unified and secure mechanism to track the movement of medicines seamlessly across all stages. Most existing systems rely on centralized databases or manual record-keeping, which leads to inefficiencies, lack of synchronization, and potential security vulnerabilities.

One of the major problems in the current system is the lack of transparency and traceability. Since data is stored in isolated or disconnected systems, stakeholders have limited visibility into the overall supply chain. This makes it difficult to verify the origin, journey, and authenticity of pharmaceutical products. As a result, counterfeit or substandard medicines can enter the market without being easily detected, posing serious health risks to patients and affecting the credibility of pharmaceutical companies.

Another significant issue is the risk of data tampering and unauthorized access. Centralized systems are vulnerable to cyberattacks, internal manipulation, and accidental data loss. Important information such as batch details, manufacturing dates, and expiry dates can be altered or misrepresented, compromising the integrity and reliability of the data. This lack of secure data management reduces trust among stakeholders and creates loopholes for fraudulent activities.

Additionally, the absence of real-time tracking and monitoring makes it difficult to keep track of product movement across different stages of the supply chain. Delays, losses, or irregularities in the system are not easily identified, leading to inefficiencies and operational challenges. In cases of defective or expired products, tracing back to the source becomes time-consuming and complicated.

Furthermore, the heavy dependence on manual processes and paper-based documentation increases the chances of human errors, duplication of records, and data

inconsistencies. The lack of proper coordination and communication among stakeholders further worsens the situation, making the entire system less reliable and inefficient.

These challenges highlight the need for a secure, transparent, and efficient solution that can ensure proper tracking, verification, and management of pharmaceutical products throughout the supply chain, thereby improving safety, trust, and overall system performance.

2.2 Motivation

The motivation for addressing this problem arises from the growing need to ensure drug safety, authenticity, and transparency in the pharmaceutical industry. In recent years, the rise of counterfeit and substandard medicines has become a serious global concern. These fake drugs not only cause financial losses to pharmaceutical companies but also pose severe health risks to patients, sometimes leading to life-threatening consequences.

Existing pharmaceutical supply chain systems lack proper mechanisms for secure data sharing and real-time visibility. Since information is often stored in separate and centralized databases, stakeholders are unable to fully verify product details or track the movement of medicines effectively. This lack of transparency creates gaps in the system, making it easier for counterfeit products to enter the market without detection.

Another key issue is the lack of trust and accountability among stakeholders. As multiple parties are involved, there is no single reliable source of truth that all participants can depend on. This leads to inefficiencies, delays in verification, and increased chances of fraud or mismanagement. End consumers, who ultimately rely on the safety and effectiveness of medicines, are the most affected by these shortcomings.

Furthermore, the absence of a tamper-proof system for maintaining records increases the risk of data manipulation and unauthorized changes. This reduces confidence in the system and makes it difficult to ensure the integrity of critical information such as product origin, batch details, and expiry dates.

Therefore, this project is motivated by the need to:

- Eliminate the presence of counterfeit drugs in the supply chain
- Ensure secure, transparent, and tamper-proof data management
- Improve visibility and coordination among all stakeholders
- Enable real-time tracking and easy verification of pharmaceutical products

By leveraging blockchain technology, the proposed system aims to provide a decentralized and trustworthy solution where all transactions are recorded securely and cannot be altered. This approach enhances reliability, builds trust among stakeholders, and improves the overall efficiency of the pharmaceutical supply chain.

2.3 Objectives

The main objective of this project is to develop a secure, transparent, and efficient blockchain-based system that addresses the existing challenges in the pharmaceutical supply chain. The system aims to ensure reliable tracking, verification, and management of medicines, thereby improving overall trust and safety in the healthcare ecosystem.

To achieve this, the project focuses on the following specific objectives:

- To design and implement a system that enables end-to-end tracking of medicines, covering every stage from manufacturing to final delivery
- To utilize blockchain technology to create a decentralized platform that eliminates the risks associated with centralized systems, such as data tampering and single points of failure
- To ensure data integrity and immutability, where all records once stored cannot be altered or deleted, thereby maintaining a trustworthy system

- To implement smart contracts for automating key operations such as batch creation, ownership transfer, and validation of transactions, reducing manual effort and human errors
- To provide a user-friendly interface that allows stakeholders to easily interact with the system, monitor product movement, and verify details in real time
- To enhance transparency, trust, and accountability among all participants by maintaining a shared and verifiable record of transactions
- To minimize and prevent the entry of counterfeit medicines into the supply chain by enabling quick and reliable authenticity checks

In addition to these functional goals, the project also aims to demonstrate the practical application of modern technologies like blockchain in solving real-world problems in the pharmaceutical sector. By ensuring that every transaction is secure, transparent, and traceable, the system contributes to building a more reliable and efficient supply chain.

Chapter 3

Detailed Survey

This chapter presents a concise review of twenty research papers relevant to the proposed blockchain-based pharmaceutical supply chain system. The papers cover key areas such as drug traceability, anti-counterfeit mechanisms, smart contracts, IoT integration, and supply chain transparency. Each paper is summarised focusing on the methodology adopted, key observations, and notable results, so as to highlight research gaps that the proposed system aims to address.

In [1], the study proposes a blockchain-based healthcare supply chain framework integrated with machine learning for detecting and managing unknown viral diseases. The system uses blockchain to provide secure, decentralized, and tamper-proof storage of healthcare records, while machine learning models perform symptom clustering and anomaly detection. Smart contracts and Zero-Knowledge Proofs are incorporated to ensure secure validation and privacy preservation of patient information. Geospatial analysis is also used to monitor disease spread and predict medicine demand in different regions. The framework achieved 87% accuracy in disease-category classification and improved outbreak tracking and medicine shortage prediction.

The study in [2] focuses on improving pharmaceutical supply chain transparency and efficiency using blockchain technology. A combination of literature review, case study analysis, and prototype implementation was used to trace medicine flow from manufacturers to pharmacies through decentralized ledgers and smart contracts. The proposed system ensured secure and tamper-resistant storage of transaction records across the network. Smart contracts automated order approvals and payment settlements, reducing operational delays and human errors. Real-time drug tracking improved inventory management and minimized fraud risks, while maintaining transparency and trust among all supply chain participants.

Here in [3], a Hyperledger Fabric-based blockchain framework was developed to secure and trace pharmaceutical supply chain activities. All stakeholders operated as nodes in a decentralized network, where every transaction related to drug production, shipment, and

delivery was recorded and validated through smart contracts. The survey in [3] observed that traditional pharmaceutical systems are vulnerable to counterfeit drugs and data manipulation. By implementing blockchain, the framework enabled secure real-time tracking and transparent record sharing among participants. The system significantly improved drug traceability, prevented unauthorized data modifications, and allowed consumers to verify medicine authenticity directly from the ledger.

According to [4], counterfeit and substandard pediatric medicines remain a major healthcare challenge in low- and middle-income countries due to weak regulatory monitoring and poor-quality control. The study conducted a narrative review using scientific databases, pharmaceutical studies, and policy reports to analyze the causes and distribution of falsified medicines. The proposed system highlighted that illegal drug markets and limited governance contribute significantly to medicine-related risks. It also emphasized the importance of collaboration among healthcare organizations, regulatory authorities, and pharmaceutical stakeholders. Strengthening monitoring systems, improving reporting mechanisms, and enforcing international regulations were identified as effective strategies to reduce counterfeit medicines globally.

As seen in [5], counterfeit medicines continue to pose a serious public health threat worldwide, leading to treatment failure, adverse reactions, and drug resistance. The study reviewed traditional analytical methods along with emerging technologies such as Blockchain, Artificial Intelligence, RFID, QR codes, and hologram-based authentication systems. The survey in this reported that nearly 10% of global medicines are falsified, with counterfeit trafficking increasing rapidly across multiple countries. The proposed framework suggested that combining blockchain with AI provides a powerful solution for ensuring supply chain transparency and real-time anomaly detection. Additionally, anti-counterfeit packaging technologies further strengthened medicine authentication and supply chain security.

The study in [6] examines the role of blockchain technology in improving transparency within agri-food supply chains through thematic analysis of research studies and industry reports. The framework highlighted blockchain features such as traceability,

immutability, and smart contracts, especially when integrated with IoT sensors for real-time monitoring. The survey in [6] observed that blockchain enables secure data sharing among farmers, processors, distributors, and retailers, improving product visibility from source to consumer. Integration with IoT enhanced trust by ensuring data accuracy and tamper resistance. The findings demonstrated that blockchain reduces fraud and strengthens traceability, which can also be effectively applied to pharmaceutical supply chain management.

Here in [7], a Fused Multi-Scale Metric (FMSM) model was proposed to identify critical nodes in multi-layer supply chain networks. The methodology combined several centrality measures with ESG performance indicators to analyse complex supply chain structures using real industrial data. The study observed that certain nodes strongly influence resource coordination, information flow, and network stability within supply chains. The proposed system provided a more comprehensive understanding of supply chain behaviour compared to traditional single-metric approaches. Results identified 35 important enterprises with strong governance and social responsibility performance, offering useful insights for identifying key stakeholders in pharmaceutical supply chain systems.

According to [8], blockchain technology can significantly improve supply chain resilience through transparency, collaboration, trust, and real-time information sharing. The study used a systematic literature review along with DEMATEL analysis to identify major enablers and barriers affecting blockchain adoption. The survey in [8] revealed that technical complexity, lack of standardization, and limited stakeholder acceptance are the primary obstacles to implementation. However, transparency and collaboration were identified as the most important enabling factors. The study concluded that establishing industry-wide standards and improving stakeholder awareness are essential for successful blockchain adoption in pharmaceutical supply chains.

As seen in [9], traditional supply chains often face poor coordination and isolated data management systems, resulting in low transparency and inefficient communication. The proposed blockchain-based information sharing mechanism used a five-layer architecture to support secure and transparent data exchange among suppliers, manufacturers, and sellers. Blockchain improved coordination by allowing all stakeholders to access shared and verified

records in real time. The framework enhanced information transparency, traceability, and operational efficiency across the network. The results confirmed blockchain as an effective technology for secure information sharing in complex supply chain environments.

The study in [10] presents a blockchain-based pharmaceutical supply chain management system using Ethereum and IPFS to prevent drug counterfeiting. Smart contracts automated supply chain activities, while digital signatures and Proof of Work consensus ensured secure transaction validation. The proposed system addressed the lack of transparency in traditional drug supply chains by enabling secure and visible product tracking among manufacturers, distributors, and retailers. Results showed improvements in scalability, throughput, reliability, and traceability compared to centralized systems. Integration of IPFS with Ethereum also improved large-scale data storage without increasing blockchain overload.

Here in [11], a Hyperledger Fabric-based blockchain framework integrated with IoT devices was proposed for medicine anti-counterfeiting and secure supply chain management. Multi-level authentication using encrypted QR codes and blockchain watermarking enabled secure medicine verification and shipment tracking across all stakeholders. The study observed that combining blockchain with IoT significantly improved real-time verification and transparency throughout the pharmaceutical supply chain. The proposed system was highly effective in detecting counterfeit medicines within distributed environments. Simulation results demonstrated high scalability, lower latency, and throughput of 417.5 transactions per second, making the framework suitable for large-scale deployment.

According to [12], blockchain combined with homomorphic encryption can provide a secure and privacy-preserving pharmaceutical supply chain system. The framework used Ethereum, IPFS, and smart contracts to manage transactions while encrypted verification ensured secure tracking of medicines across all distribution stages. The survey in [12] highlighted that traditional supply chains often lack proper transparency and secure stakeholder coordination. Homomorphic encryption allowed data verification without exposing sensitive information, while blockchain maintained tamper-proof traceability. The system achieved nearly 98% accuracy in counterfeit tag detection and significantly improved medicine authentication and supply chain security.

As seen in [13], blockchain integrated with reinforcement learning can create a smarter pharmaceutical supply chain management system. Blockchain ensured secure and transparent storage of transaction data, while reinforcement learning analysed patient reviews and ratings to recommend suitable medications. The study observed that the integration improved both traceability and personalized decision-making in drug distribution. The proposed framework was evaluated using more than 215,000 drug reviews and achieved around 92% recommendation accuracy. The results demonstrated the practical benefits of combining blockchain and machine learning technologies in healthcare applications.

The study in [14] proposes a blockchain-integrated secure supply chain framework with multiple authentication and detection algorithms to ensure product quality and counterfeit tag detection. The system included reliable manufacturer selection, sensor-based smart traceability, and algorithmic verification mechanisms for identifying counterfeit products. The survey in [14] observed that combining blockchain with sensor tracking significantly improved transparency and prevented poor-quality products from entering the supply chain. The framework provided complete visibility of product movement at every stage. Experimental evaluation achieved nearly 98% accuracy in counterfeit detection and authenticity verification while maintaining efficient system performance.

Here in [15], a Natural Resource-Based View (NRBV) approach was used to study circular economy practices in pharmaceutical industries through multiple case studies and semi-structured interviews. The study identified practices such as green procurement, sustainable production, reverse logistics, and digitalization as important factors for reducing environmental impact and improving operational efficiency. The proposed system emphasized the role of digital technologies in supporting sustainable pharmaceutical supply chain operations. Results showed approximately 95% secure transaction reliability along with improved blockchain-based traceability and data privacy. The study concluded that integrating blockchain with circular economy practices enhances both sustainability and supply chain security.

The study in [16] proposes a blockchain-based drug traceability framework that tracks medicines from manufacturers to consumers using a decentralized ledger system. Each stakeholder including manufacturers, distributors, pharmacies, and regulators records

transactions on the blockchain to maintain a complete and verifiable product history. The survey in this observed that centralized pharmaceutical databases are vulnerable to data tampering, weak authentication, and lack of transparency. Blockchain technology addressed these issues by enabling secure and distributed transaction verification among authorized participants. The framework improved medicine authenticity verification, secure tracking, and transparency while allowing consumers to validate drugs before purchase, thereby reducing counterfeit medicine risks.

Here in [17], an integrated engineering and IT management framework was developed using blockchain-enabled traceability for pharmaceutical supply chains. The proposed system combined IoT tracking devices, blockchain storage, and smart contracts to monitor pharmaceutical products throughout manufacturing and healthcare delivery processes. The study observed that poor coordination between engineering systems and IT infrastructure often causes delays in identifying counterfeit products and supply chain disruptions. Integrating blockchain with IoT significantly improved operational transparency, regulatory compliance, and secure data sharing among stakeholders. Results demonstrated enhanced supply chain resilience, faster product recalls, improved monitoring efficiency, and increased stakeholder trust, especially during emergency situations such as pandemics.

According to [18], a blockchain-based framework integrated with the Edwards-curve Digital Signature Algorithm (EdDSA) was proposed to strengthen pharmaceutical supply chain security. Each medicine unit was assigned a unique QR code connected to blockchain records, allowing stakeholders to verify product authenticity at every stage of the supply chain. The survey in this highlighted that traditional cryptographic methods often involve higher computational costs and slower transaction processing. By using EdDSA digital signatures, the framework improved efficiency while maintaining strong security and privacy protection. The system achieved reduced computational overhead and enabled instant medicine verification through QR-code-based authentication, making it practical for real-world pharmaceutical applications.

As seen in [19], an integrated IoT-Cloud-Blockchain architecture was introduced to improve pharmaceutical supply chain management and logistics monitoring. IoT sensors continuously monitored environmental conditions such as temperature and humidity during

transportation, while blockchain securely stored sensor readings and transaction data in a tamper-proof manner. The study observed that pharmaceutical logistics systems commonly face issues including fragmented data management, limited real-time monitoring, and security vulnerabilities. Combining IoT, cloud computing, and blockchain significantly improved automated monitoring and ensured secure real-time accessibility of logistics information. Experimental evaluation confirmed enhanced traceability, communication efficiency, operational performance, and data security throughout the pharmaceutical supply chain.

The study in [20] presents a blockchain-based pharmaceutical supply chain network where manufacturers, distributors, regulators, and pharmacies operate as nodes within a decentralized environment. Smart contracts were used to automate supply chain transactions according to predefined conditions without requiring manual intervention. The proposed framework addressed the limitations of traditional centralized systems, which often suffer from delays, inefficient coordination, and risks of data manipulation. Blockchain technology enabled secure data sharing and transparent transaction verification among all participants in the network. Results demonstrated improved operational efficiency, enhanced collaboration, secure tracking of pharmaceutical products, and effective prevention of counterfeit medicines entering the market.

Chapter 4

Survey Summary Table

This chapter provides summary of the reviewed papers as given in Table 4.1

Table 4.1 Summary of survey papers

Sl. no	Title of the Paper	Problem Addressed	Method / Approach	Results
1	A Robust Blockchain-Based Framework for Managing Unknown Viral Diseases (2025)	Lack of secure disease outbreak detection and medicine demand prediction in healthcare	Blockchain + ML (Naive Bayes, Gradient Boosting) + Zero-Knowledge Proofs + geospatial analysis	87% disease classification accuracy; effective unknown disease detection via anomaly detection
2	Implementing Blockchain for Improved Pharmaceutical Transparency and Efficiency (2024)	Lack of transparency, fraud, and inefficiency in pharmaceutical supply chains	Literature review + case study + blockchain prototype with smart contracts and decentralized ledger	Improved security, traceability, and regulatory compliance; reduced counterfeit drug risks
3	A Blockchain Driven Framework for Pharmaceutical Drug Supply Chain (2021)	Counterfeit drugs and poor data transparency in drug distribution	Hyperledger Fabric blockchain with smart contracts and decentralized stakeholder network	Improved drug traceability, transparency, and supply chain security;

4	Substandard and Falsified Pediatric Medicines in Low-Middle-Income Countries (2025)	High prevalence of counterfeit medicines in developing countries with weak regulatory systems	Narrative literature review of scientific databases, policy reports, and pharmaceutical studies	Strengthened regulatory enforcement and global cooperation can significantly reduce falsified medicines
5	Counterfeit Medicine: A Major Public Health Concern (2025)	Rising global prevalence of counterfeit drugs causing treatment failure and death	Comprehensive review of detection techniques (HPLC, Blockchain, AI, QR codes, RFID, holograms)	Blockchain + AI identified as most effective defense; multi-layer authentication reduces counterfeiting
6	Blockchain Technology for Transparency in Agri-Food Supply Chain (2024)	Lack of transparency and product traceability in food and supply chain networks	Thematic analysis of research studies + IoT sensor integration with blockchain smart contracts	Multiple blockchain use cases identified; improved consumer trust and product traceability
7	Examining Factors That Facilitate or Hinder Blockchain Adoption in Supply Chains (2024)	Unclear enablers and barriers to blockchain adoption for supply chain resilience	Systematic literature review + DEMATEL analysis + expert opinions from multiple industries	11 enablers and 12 barriers identified; transparency and trust are top enablers

8	Blockchain-Based Supply Chain Information Sharing Mechanism (2022)	Poor information sharing and isolated data systems in supply chains	Five-layer blockchain architecture for managing data exchange and coordination among participants	Improved information transparency, traceability, and security; enhanced stakeholder collaboration
9	Blockchain-Based Secure and Scalable System to Prevent Drug Counterfeiting (2024)	Counterfeit drugs and lack of transparency in traditional drug supply chains	Ethereum + IPFS + smart contracts + digital signatures + Proof of Work consensus	Improved drug traceability, security, and scalability; better throughput than existing systems
10	A Multilevel Authentication-Based Blockchain for IoT Supply Chain (2024)	Counterfeit medicines and insecure data sharing in IoT-enabled supply chains	Hyperledger Fabric + IoT + encrypted QR codes + multi-level blockchain	417.5 TPS with 100,000 transactions; improved throughput, lower latency
11	Blockchain and Homomorphic Encryption Against Counterfeit Drugs (2025)	Lack of privacy-preserving transparency in pharmaceutical supply chain systems	Ethereum + IPFS + homomorphic encryption + smart contracts for secure drug tracking	98% counterfeit tag detection accuracy; improved authentication and product traceability
12	Supply Chain Management	Poor drug distribution	Blockchain for secure records +	92% recommendation

	Using Blockchain and Reinforcement Learning (2023)	management and lack of personalized medication recommendations	reinforcement learning model trained on drug review dataset	accuracy on 215,000+ drug reviews; improved decision support for distribution
13	Drug Traceability Using Blockchain (2024)	Data tampering and weak authentication in centralized pharmaceutical databases	Decentralized blockchain ledger with all stakeholders recording transactions as network nodes	Tamper-proof records; improved drug authentication; consumers can verify medicines before purchase
14	Blockchain-Enabled Traceability: Engineering and IT Management Framework (2025)	Poor coordination between engineering systems and IT in pharmaceutical supply chains	Blockchain + IoT tracking + smart contracts + IT governance for regulatory compliance	Enhanced real-time traceability; faster product recalls; improved stakeholder trust
15	A Blockchain-Based Framework for Drug Security (2025)	High computational cost and weak authentication in blockchain supply chain systems	Ethereum blockchain + EdDSA digital signatures + QR codes linked to blockchain records	Reduced computational overhead; improved security; instant drug verification via QR code scanning, lower latency

16	Secure Pharmaceutical Supply Chain Using IoT-Cloud-Blockchain Architecture (2024)	Data fragmentation and lack of real-time monitoring in pharmaceutical logistics	IoT sensors + cloud computing + blockchain for environmental condition tracking during transport	Improved monitoring, traceability, and data security; enhanced operational efficiency in logistics
17	Blockchain-Based Pharmaceutical Supply Chain Network Using Smart Contracts (2024)	Over-reliance on intermediaries and centralized databases in pharmaceutical networks	Blockchain network where stakeholders operate as nodes; smart contracts automate all processes	Improved transparency and efficiency; prevented counterfeit entry; enhanced inter-stakeholder collaboration
18	Securing Supply Chain with Blockchain-Integrated Algorithmic System (2024)	Counterfeit product tags and poor product quality in supply chain systems	Blockchain + sensor-based smart traceability + algorithms + manufacturer selection module	98% counterfeit tag detection accuracy; improved traceability and supply chain security
19	A Natural Resource-Based View of Circular Economy in Pharmaceutical Industry (2026)	Environmental impact and inefficient resource use in pharmaceutical supply chains	Multiple case study approach using NRBV theory + 18 semi-structured manager interviews	95% secure transaction reliability; improved privacy and efficient blockchain drug traceability

20	Blockchain Technology for Transparency in Agri-Food Supply Chain (2024)	Fraud, lack of traceability, and low consumer trust in product supply chains	Thematic analysis of blockchain use cases with IoT and smart contracts in supply chains	Reduced fraud; improved product traceability; increased consumer trust and supply chain efficiency
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Chapter 5

System Requirement Specification

This chapter details various requirements to carry out our project

5.1 Functional Requirements

Functional requirements describe the essential features and operations that the system must perform to meet the objectives of the project. The proposed blockchain-based pharmaceutical supply chain system includes the following key functionalities:

- The system should allow the manufacturer to create a new medicine batch by entering all necessary details such as batch ID, medicine name, quantity, manufacturing date, and expiry date. Once created, this batch serves as the starting point for tracking the product throughout the supply chain.
- The system should enable secure recording of all batch details on the blockchain. Each entry must be stored in a tamper-proof manner, ensuring that the data cannot be modified or deleted once it is validated and added to the blockchain.
- The system should support ownership transfer of medicine batches between different stakeholders such as manufacturers, distributors, and retailers. Every transfer must be recorded as a transaction on the blockchain, ensuring transparency and accountability.
- The system should maintain a complete and chronological history of all transactions associated with each medicine batch. This allows users to trace the entire lifecycle of a product, from its origin to its current state.
- The system should provide real-time tracking capabilities, enabling users to monitor the current status and location of pharmaceutical products at any stage in the supply chain.
- The system should allow users to verify the authenticity of medicines by accessing the stored blockchain data. Users can check details such as origin, ownership history, and product information to confirm whether a medicine is genuine.

- The system should integrate with MetaMask for secure user authentication and transaction signing. This ensures that only verified users can interact with the blockchain and perform operations.
- The system should provide a user-friendly interface that allows users to easily perform actions such as creating batches, transferring ownership, and viewing product details without requiring deep technical knowledge.
- The system should enforce role-based access control, ensuring that only authorized users (e.g., manufacturers) can perform specific actions like batch creation, while others (e.g., distributors) can only perform permitted operations.
- The system should display clear feedback messages and transaction status updates for all user actions, such as successful batch creation, ownership transfer confirmation, or error messages in case of failed transactions.

5.2 Non-Functional Requirements

Non-functional requirements define the quality attributes, performance standards, and operational constraints of the system. These requirements ensure that the system not only functions correctly but also perform efficiently, securely, and reliably under different conditions.

- **Security:**

The system must ensure a high level of security for all transactions and data. By using blockchain technology and cryptographic techniques, all records are securely stored and protected from unauthorized access or modification. Integration with MetaMask ensures that only authenticated users can perform transactions, and all actions require proper digital signatures.

- **Performance:**

The system should provide efficient performance with minimal delay in processing transactions. Although blockchain transactions may take some time depending on network conditions, the system should be optimized to handle operations smoothly and provide timely feedback to users.

- **Scalability:**

The system should be capable of handling an increasing number of users and transactions as the application grows. The architecture should support future expansion, allowing integration of additional features and larger datasets without significant performance degradation.

- **Reliability:**

The system must deliver consistent and accurate results under all conditions. Since blockchain ensures data immutability and distributed storage, the system should remain reliable even in cases of partial failures or network issues.

- **Usability:**

The user interface should be simple, intuitive, and easy to navigate. Users with minimal technical knowledge should be able to perform operations such as batch creation, tracking, and verification without difficulty. Clear instructions and feedback messages should be provided.

- **Availability:**

The system should be accessible at all times with minimal downtime. Since blockchain networks are decentralized, the system benefits from high availability, ensuring that users can access data and perform transactions whenever required.

- **Maintainability:**

The system should be designed in a modular and organized manner, making it easy to update, debug, and extend. Future improvements, bug fixes, or feature additions should be implementable without affecting the overall system stability.

- **Transparency:**

All transactions recorded on the blockchain should be transparent and visible to authorized participants. This ensures trust among stakeholders, as every action can be verified and traced back when needed.

- **Data Integrity:**

The system must ensure that once data is recorded on the blockchain, it cannot be altered or deleted. This immutability guarantees the accuracy and trustworthiness of the stored information, which is critical in the pharmaceutical domain.

5.3 Hardware Requirements

The hardware requirements for the proposed system are minimal, as it is primarily a web-based application that interacts with a blockchain network. The system does not require any specialized or high-end hardware, making it accessible for development and usage on standard computing devices.

The basic hardware requirements include:

- A computer system or laptop with the following specifications:
 - o **Processor:** Intel i3 or higher (Intel i5 or equivalent recommended for better performance)
 - o **RAM:** Minimum 4 GB (8 GB recommended for smooth execution, especially during blockchain operations)
 - o **Storage:** At least 10-20 GB of free disk space for project files, dependencies, and blockchain artifacts
- A stable and reliable internet connection, which is essential for interacting with the blockchain network, deploying smart contracts, and performing transactions
- Optional hardware support:
 - o A smartphone device can be used if MetaMask mobile is integrated for extended usability
 - o External storage devices for backup purposes (optional but recommended for project safety)

These requirements ensure that the system can be developed, tested, and executed efficiently without the need for expensive infrastructure.

5.4 Software Requirements

The software requirements define the technologies, platforms, and tools used for the development, deployment, and execution of the proposed system. The system is built using

modern web and blockchain technologies to ensure efficiency, scalability, security, and ease of development.

The required software components include:

- **Operating System:** The system can be developed and executed on Windows, Linux, or macOS. Any operating system that supports Node.js and modern development tools is suitable for this project.
- **Programming Languages:**
 - o Solidity is used for writing smart contracts that define the logic of the pharmaceutical supply chain on the blockchain
 - o JavaScript is used for frontend development and for interacting with the blockchain through libraries.
- **Blockchain Platform:** Ethereum is used as the underlying blockchain platform for deploying and executing smart contracts. It provides a decentralized and secure environment for storing transaction data.
- **Development Framework and Tools:**
 - o Hardhat is used as the primary development framework for compiling, testing, and deploying smart contracts in a local blockchain environment.
 - o Node.js acts as the runtime environment for executing JavaScript code and managing backend operations.
 - o npm (Node Package Manager) is used to install and manage required libraries and dependencies efficiently.
- **Frontend Technologies:**
 - o HTML, CSS, and JavaScript are used to design and develop the user interface.
 - o (Optional) React.js can be used for building a more dynamic and responsive frontend.

- **Blockchain Interaction Libraries:** Ethers.js is used to establish communication between the frontend and the blockchain, allowing users to interact with deployed smart contracts and perform transactions.
- **Wallet Integration:** MetaMask is used for user authentication, account management, and secure transaction signing. It acts as a bridge between the browser and the Ethereum blockchain.
- **Web Browser:** Google Chrome or any modern web browser that supports the MetaMask extension is required to run and interact with the application.

These software components work together to create a secure, decentralized, and user-friendly application. They enable seamless interaction between users and the blockchain while ensuring that all operations are performed efficiently and securely within the pharmaceutical supply chain system.

Additionally, the selected software stack provides flexibility, reliability, and ease of maintenance for future development and upgrades. The combination of blockchain technology with modern web development tools ensures that the system remains scalable, secure, and efficient for handling pharmaceutical supply chain operations.

The integration of technologies such as Ethereum, Hardhat, Ethers.js, and MetaMask enables seamless communication between users and the blockchain network while maintaining transparency and data integrity. Overall, these software requirements create a strong technical foundation for developing a practical and real-world blockchain-based pharmaceutical supply chain management system.

Chapter 6

System Design

6.1 System Design

System design defines the overall structure and organization of the proposed blockchain-based pharmaceutical supply chain management system. It describes how different components of the system are arranged and how they interact with each other to achieve the desired functionality.

The system is designed using a decentralized architecture, where the core logic is implemented through smart contracts deployed on the blockchain. The frontend application acts as an interface for users to interact with the system, while blockchain technology ensures secure and immutable data storage.

The design focuses on achieving key objectives such as transparency, traceability, security, and efficiency. Each component of the system is developed in a modular manner, allowing easy maintenance, scalability, and future enhancements. The interaction between the user interface, blockchain network, and smart contracts ensures that all operations are performed securely and reliably.

6.1.1 System Architecture

The system architecture represents the overall structural design of the proposed blockchain-based pharmaceutical supply chain management system. It defines how different components of the system are organized and how they interact with each other to perform various operations efficiently. The architecture is designed to ensure security, transparency, scalability, and seamless communication between all layers of the system.

The proposed system follows a layered architecture, which consists of three main layers: the User Interface Layer, the Application Layer, and the Blockchain Layer. Each layer has a specific role and works together to provide a complete and functional system.

At the top level is the User Interface Layer, where users such as manufacturers, distributors, and retailers interact with the system. This layer is implemented using a web-

based frontend developed with technologies like HTML, CSS, and JavaScript (or React). It provides an easy-to-use interface that allows users to perform operations such as creating medicine batches, transferring ownership, and tracking product details. The design ensures that even users with minimal technical knowledge can use the system effectively.

The next layer is the Application Layer, which acts as a bridge between the frontend and the blockchain. This layer handles communication and data processing. The frontend connects to the blockchain using Ethers.js, a JavaScript library that enables interaction with smart contracts. It allows the application to send requests, retrieve data, and execute functions defined in the smart contracts.

For secure authentication and transaction approval, MetaMask is integrated into the system. MetaMask acts as a digital wallet and gateway to the blockchain network. It allows users to connect their accounts, sign transactions, and securely interact with the system. Every transaction initiated by the user must be approved through MetaMask, ensuring that only authorized actions are performed.

The Blockchain Layer forms the core of the system. This layer consists of smart contracts deployed on the Ethereum blockchain (using a local network like Hardhat during development). The smart contracts contain the business logic of the application, including functions for batch creation, ownership transfer, and data storage. These contracts automatically execute predefined rules, ensuring consistency and eliminating the need for intermediaries. All data related to medicine batches and transactions is stored on the blockchain in a decentralized and immutable manner. Once data is recorded, it cannot be altered or deleted, ensuring high data integrity and security. Each transaction is validated and added to the blockchain, creating a permanent and transparent record that can be accessed and verified by authorized users.

Figure 6.1 illustrates the overall architecture of the blockchain-based paracetamol supply chain system. The system uses role-based React dashboards for manufacturers, transporters, pharmacies, and consumers, connected through MetaMask and Web3.js. Smart

contracts developed in Solidity are deployed on the Ethereum Ganache network to store ownership and transaction details securely. This architecture ensures transparency, traceability, and tamper-proof management throughout the supply chain.

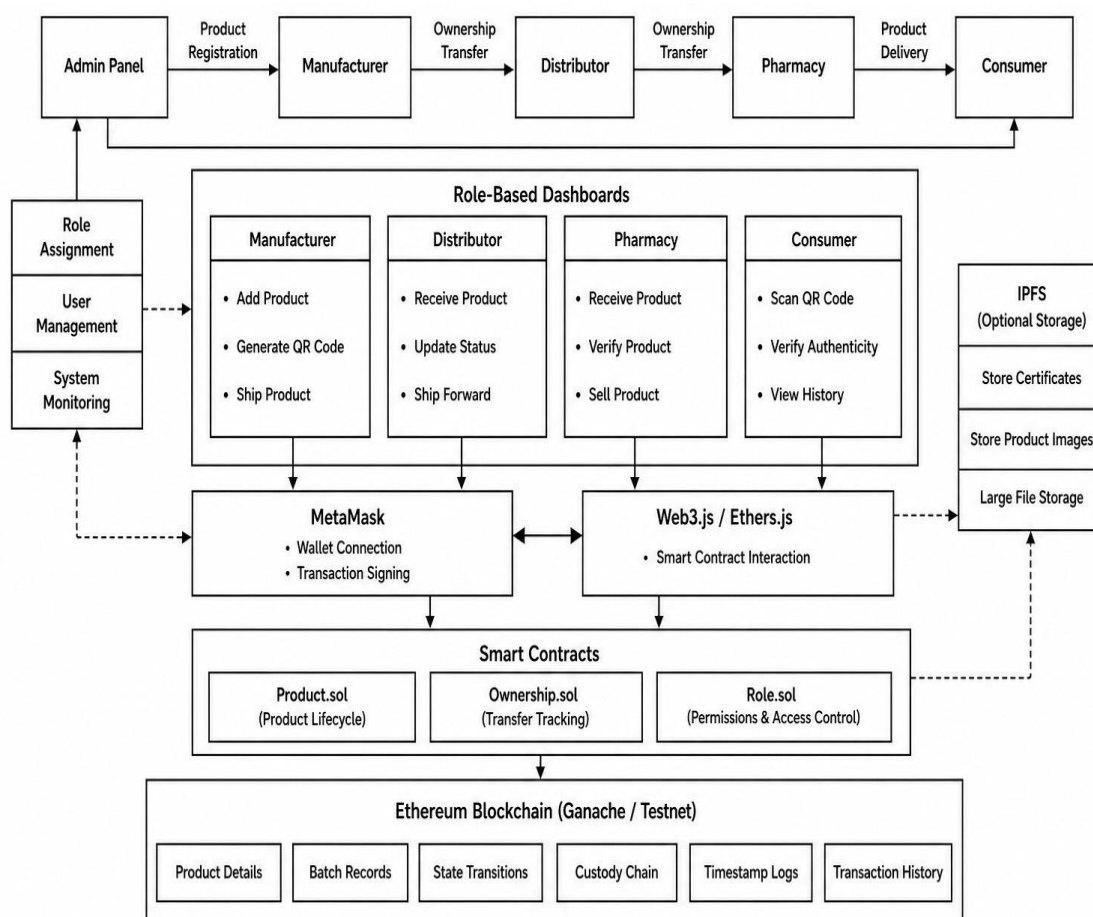


Figure 6.1 System Architecture

The overall flow of the system begins when a user performs an action through the frontend interface. The request is then processed through the application layer using Ethers.js and MetaMask, which interacts with the smart contract. The smart contract executes the requested function and records the transaction on the blockchain. The result is then sent back to the frontend and displayed to the user.

6.1.2 Module Design

The proposed system is divided into multiple modules, each responsible for handling specific functionalities within the pharmaceutical supply chain. This modular design approach ensures better organization, easier development, improved maintainability, and flexibility for future enhancements. Each module works independently while also interacting with other modules to achieve the overall system objectives.

Batch Creation Module

The Batch Creation Module is responsible for initiating the supply chain process. This module allows the manufacturer to create a new medicine batch by entering essential details such as batch ID, medicine name, quantity, manufacturing date, and expiry date. Once the data is entered, it is passed to the smart contract, which validates and stores the information on the blockchain. Since the data is recorded on the blockchain, it becomes immutable and tamper-proof, ensuring that the original details of the medicine batch cannot be altered. This module plays a crucial role in establishing the authenticity and origin of pharmaceutical products, forming the foundation for further tracking and verification.

Ownership Transfer Module

The Ownership Transfer Module manages the transfer of medicine batches between different stakeholders in the supply chain, such as distributors, wholesalers, and retailers. This module allows authorized users to initiate ownership transfers through the system interface. Each transfer is executed through a smart contract function and recorded as a blockchain transaction, ensuring transparency and accountability. The module ensures that ownership is transferred only when valid conditions are met, preventing unauthorized or fraudulent transfers. By maintaining a clear record of ownership changes, this module helps in building trust among stakeholders.

Tracking and Verification Module

The Tracking and Verification Module enables users to monitor and verify the complete lifecycle of a medicine batch. It retrieves data from the blockchain and displays detailed information such as product origin, batch details, ownership history, and current holder. This

module allows stakeholders and even ends users to verify the authenticity of medicines by checking their recorded data. It ensures that any discrepancies or irregularities in the supply chain can be easily identified. This feature is particularly important in preventing counterfeit drugs and ensuring patient safety.

User Interface Module

The User Interface Module provides an interactive and user-friendly platform for users to interact with the system. It is developed using web technologies such as HTML, CSS, and JavaScript (or react), ensuring ease of use and accessibility. This module allows users to perform operations such as batch creation, ownership transfer, and product tracking without requiring technical knowledge of blockchain. It also displays transaction results, status updates, and relevant messages, enhancing the overall user experience. The interface is designed to be intuitive, ensuring smooth navigation and usability.

Authentication Module

The Authentication Module ensures secure access to the system by integrating MetaMask, a blockchain-based wallet. This module allows users to connect their accounts and authenticate themselves before performing any operations. All transactions initiated by users must be approved through MetaMask, which provides secure transaction signing using cryptographic keys. This ensures that only authorized users can execute actions such as creating batches or transferring ownership. The module enhances system security and prevents unauthorized access or misuse.

Smart Contract Module

The Smart Contract Module forms the core of the system and contains all the business logic required for the application. It is implemented using Solidity and deployed on the Ethereum blockchain. This module defines functions for operations such as batch creation, ownership transfer, and data retrieval. It ensures that all rules and conditions are enforced automatically without the need for intermediaries. Since smart contracts are executed on the blockchain, they provide security, transparency, and reliability.

Figures 6.2 and 6.3 illustrate the data flow and internal functioning of the blockchain-based paracetamol supply chain system. Figure 6.2 shows how external stakeholders such as Admins, Manufacturers, Transporters, Pharmacies, and Consumers interact with the system for role management, batch registration, ownership transfer, and product verification. Figure 6.3 explains the internal blockchain processes where transaction records and ownership details are securely stored in the blockchain ledger. Together, these figures demonstrate a transparent, traceable, and tamper-proof supply chain management system.

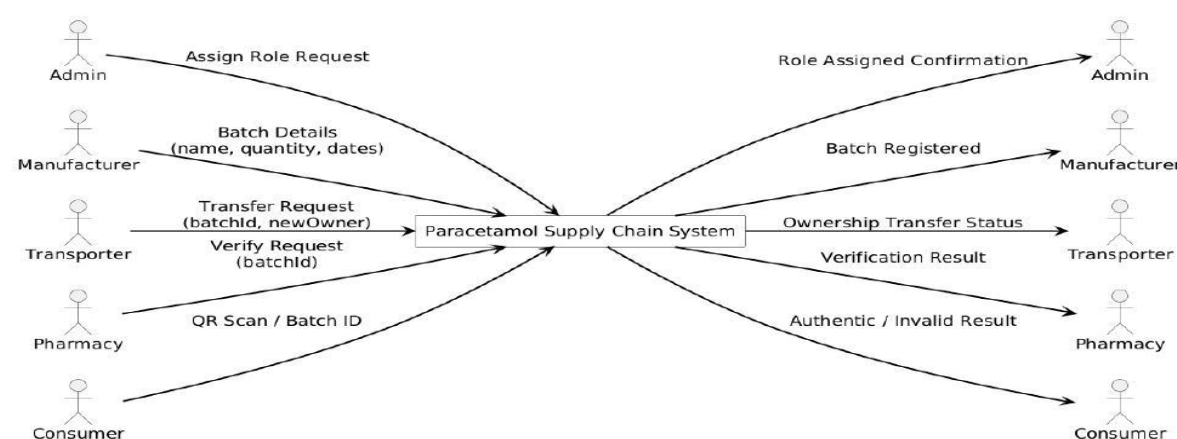


Figure 6.2 Data Flow (Level 0)

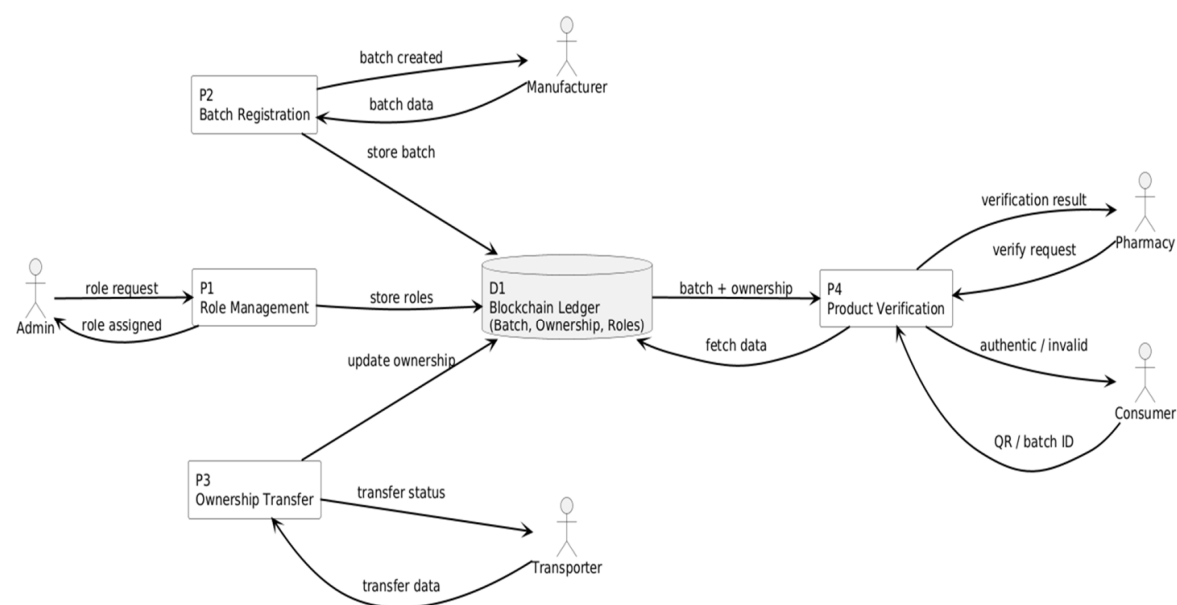


Figure 6.3 Data Flow (Level 1)

6.4 Component/Module Diagram

Figure 6.4 presents the modular architecture of the blockchain-based paracetamol supply chain system, divided into Presentation, Application, and Blockchain layers. The Presentation Layer includes role-specific interfaces for manufacturers, transporters, pharmacies, and consumers, authenticated using MetaMask. The Application Layer, built with Node.js, Express, and Web3.js, processes requests through validation and role-based access control mechanisms. The Blockchain Layer stores all verified transactions and ownership records securely using Solidity smart contracts deployed on the Ethereum Ganache network.

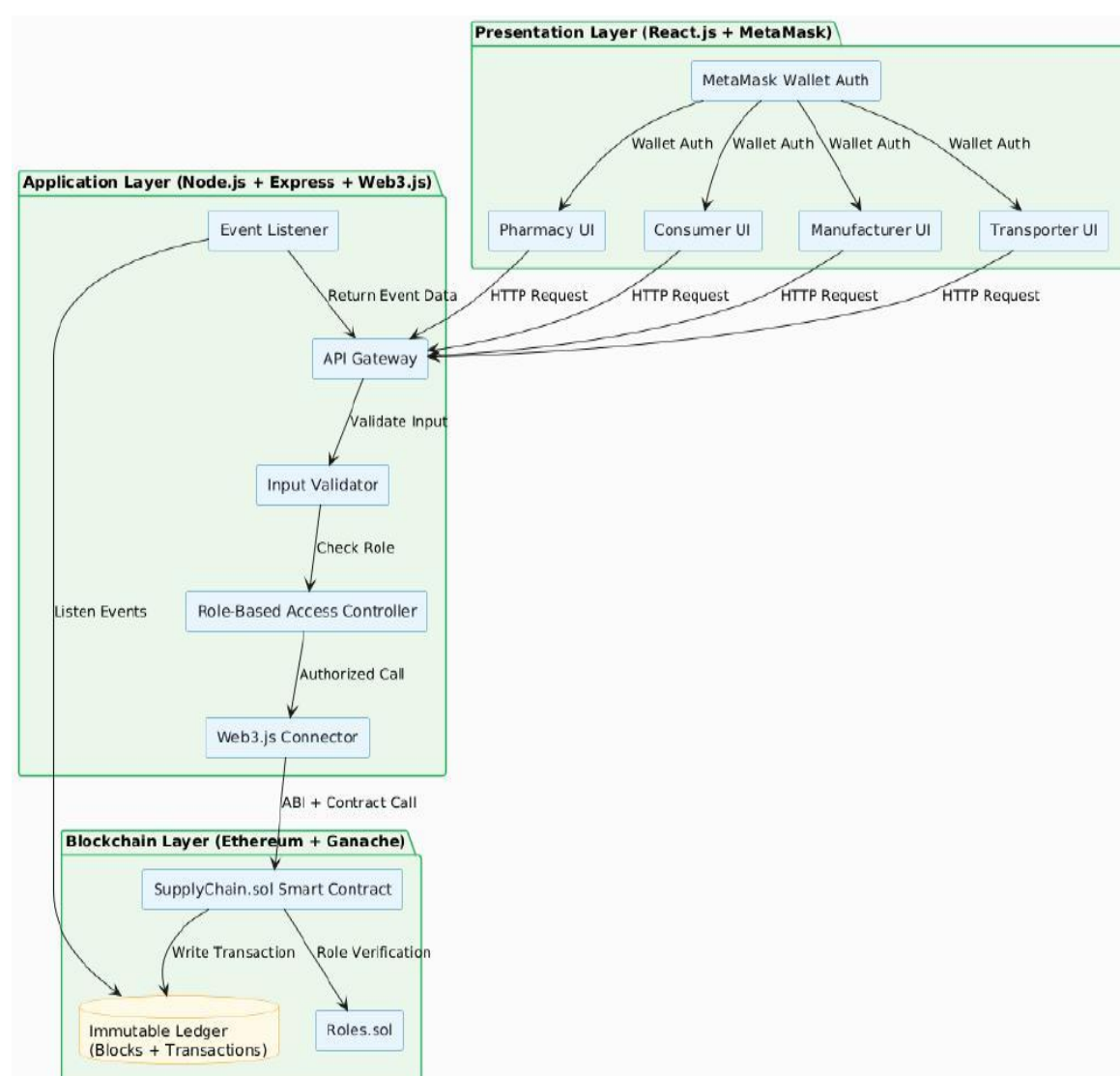


Figure 6.4 Module/Component Diagram

6.2 Detailed Design

The detailed design of the proposed Blockchain-Based Paracetamol Supply Chain Management System focuses on defining the internal structure and interaction of system components to ensure secure and efficient operation. The system is built using Ethereum blockchain technology, where smart contracts written in Solidity handle core functionalities such as batch creation, ownership transfer, and verification. Each stakeholder—including manufacturer, distributor, pharmacy, and consumer—interacts with the system through a role-based interface. The frontend is developed using a web-based framework integrated with Web3.js, which enables communication between the user interface and the blockchain network via MetaMask. The blockchain ledger maintains immutable records of all transactions, ensuring data integrity and preventing unauthorized modifications.

At the component level, the system is divided into modules such as user management, batch registration, transaction processing, and verification. The class design defines entities like Batch, User, and Transaction, each with specific attributes and methods to support system operations. The sequence of interactions begins with batch creation by the manufacturer, followed by ownership transfer through distributors to pharmacies, and finally verification by consumers using QR codes.

Each action triggers a smart contract function that records the event on the blockchain. This modular and layered design improves system scalability, enhances transparency across the supply chain, and ensures that every transaction is traceable and tamper-proof, thereby effectively addressing the issue of counterfeit medicines.

6.2.1 Class Diagram

Figure 6.5 presents the class diagram of the blockchain-based supply chain system, consisting of classes such as SmartContract, Batch, Ownership, Transaction, and User. These classes manage product details, ownership transfers, transaction records, and user roles within the system. The design ensures organized data management and secure role-based access control.

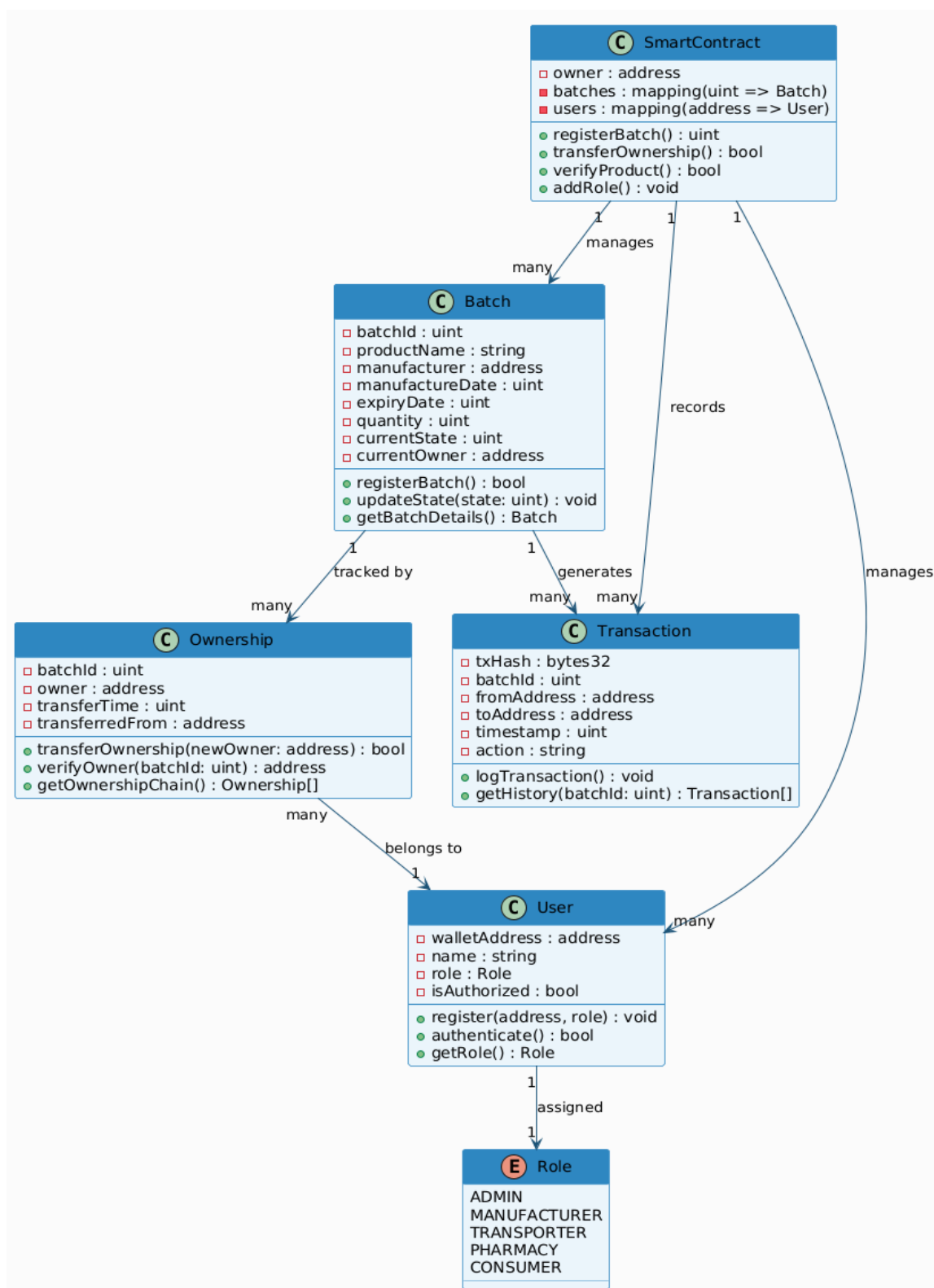


Figure 6.5 Class Diagram

6.2.2 Activity Diagram

The activity diagram captures the step-by-step workflow of the system for different stakeholder roles. The admin connects their meta mask wallet, assigns roles to addresses, and stores the role mapping on the blockchain. The manufacturer fills in a batch registration form, signs the transaction, and if the role is verified by the smart contract, the batch is stored on-chain and a QR code is generated.

The transporter enters a batch id and new owner address, and upon verification of current ownership, calls the `Transferownership()` function to update the batch status to in transit. the pharmacy verifies product authenticity using a qr scan, accepts or rejects the batch, and marks it as delivered if valid. Finally, the consumer scans the qr code and receives an authentic or do not use result along with the full ownership history. These flows demonstrate how smart contract and activity diagrams of our project logic governs every action in the system.

The Manufacturer and Transporter activities focus on registering medicine batches, generating QR codes, and securely transferring ownership during product movement. The Pharmacy and Consumer activities involve verifying medicine authenticity using batch IDs or QR codes before dispensing or purchasing products. These processes help prevent counterfeit medicines and maintain trusted tracking throughout the pharmaceutical supply chain.

Figure 6.6 illustrates the activity diagrams of different stakeholders involved in the blockchain-based pharmaceutical supply chain system, including manufacturers, transporters, pharmacies, and consumers. Each participant connects through MetaMask and performs specific operations such as batch registration, ownership transfer, and product verification. Smart contracts validate every activity to ensure that only authorized users can access and update the system. This workflow improves transparency, security, and traceability across the supply chain.

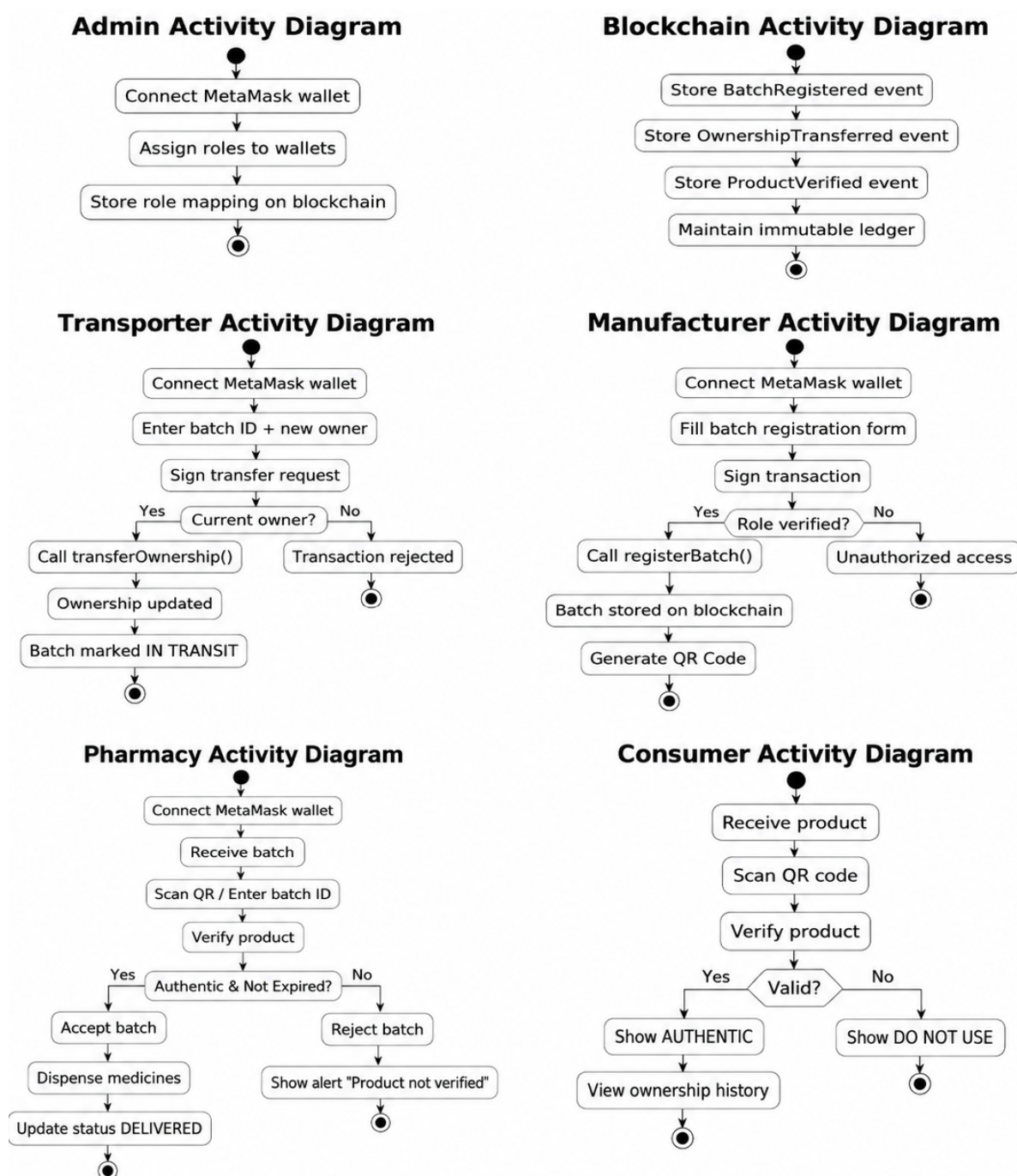


Figure 6.6 Activity Diagrams

Additionally, the activity diagram demonstrates the smooth interaction between users, the blockchain network, and smart contracts, ensuring efficient and tamper-proof management of pharmaceutical supply chain operations.

6.2.3 Use Case Diagram

Figure 6.7 presents the use case diagram of the blockchain-based paracetamol supply chain system involving Admin, Manufacturer, Transporter, Pharmacy, and Consumer roles. The system supports batch registration, ownership transfer, and QR code-based product verification through smart contracts. Admins manage user roles and monitor transactions to ensure secure and transparent supply chain operations.

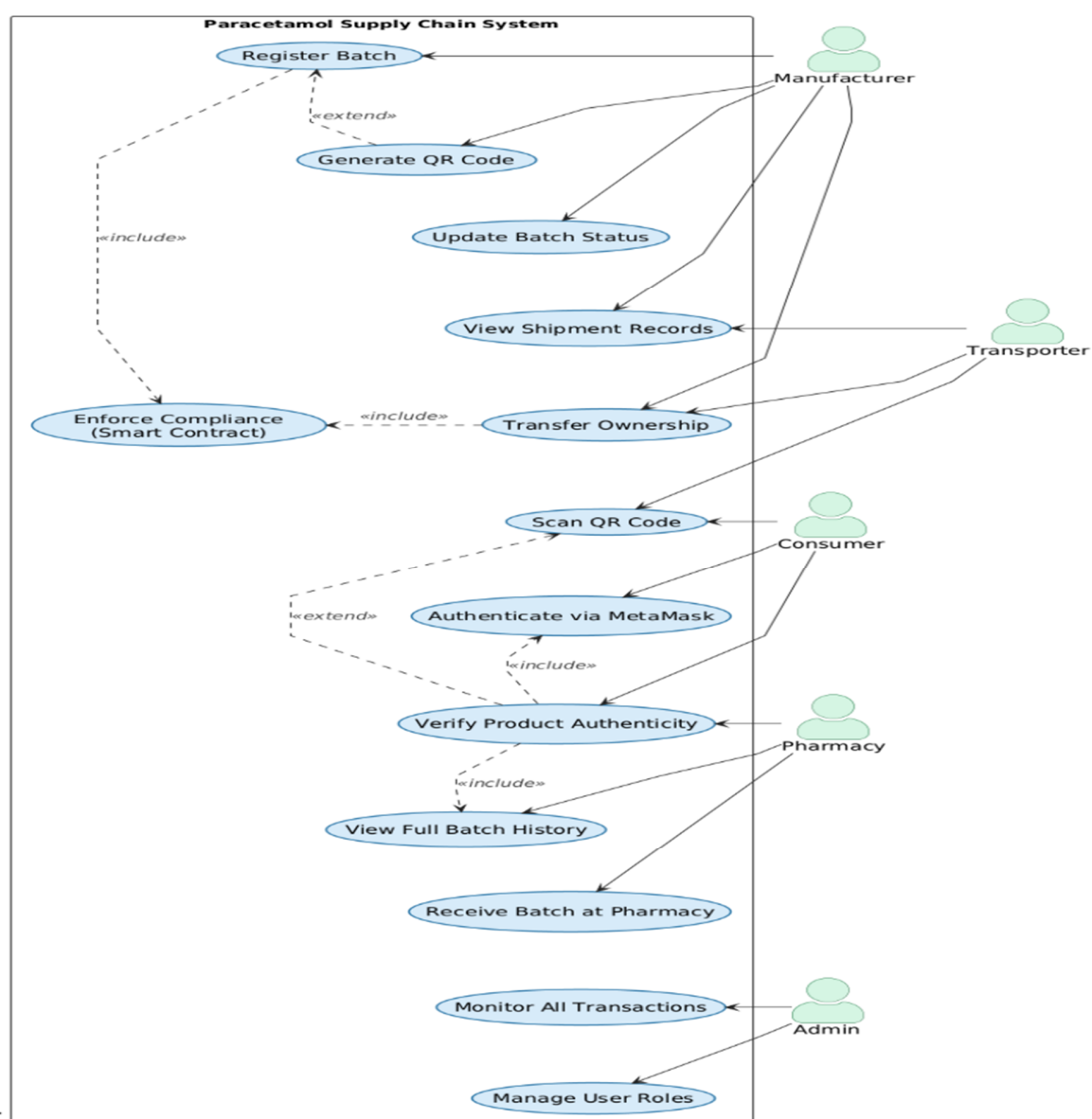


Figure 6.7 Use Case Diagram

6.2.4 Sequence Diagrams

Batch Creation

Figure 6.8 illustrates the sequence diagram for batch creation in the blockchain-based pharmaceutical supply chain system. The Manufacturer enters batch details through the React UI and signs the request using MetaMask before it is sent to the API Gateway. The system validates the input and verifies the manufacturer role through the RBAC Controller before forwarding the request to the smart contract via the Web3.js Connector. The smart contract registers the batch, stores transaction details on the Ethereum ledger, emits a BatchRegistered event, and returns a success response containing the batchId and txHash, ensuring secure and immutable batch registration.

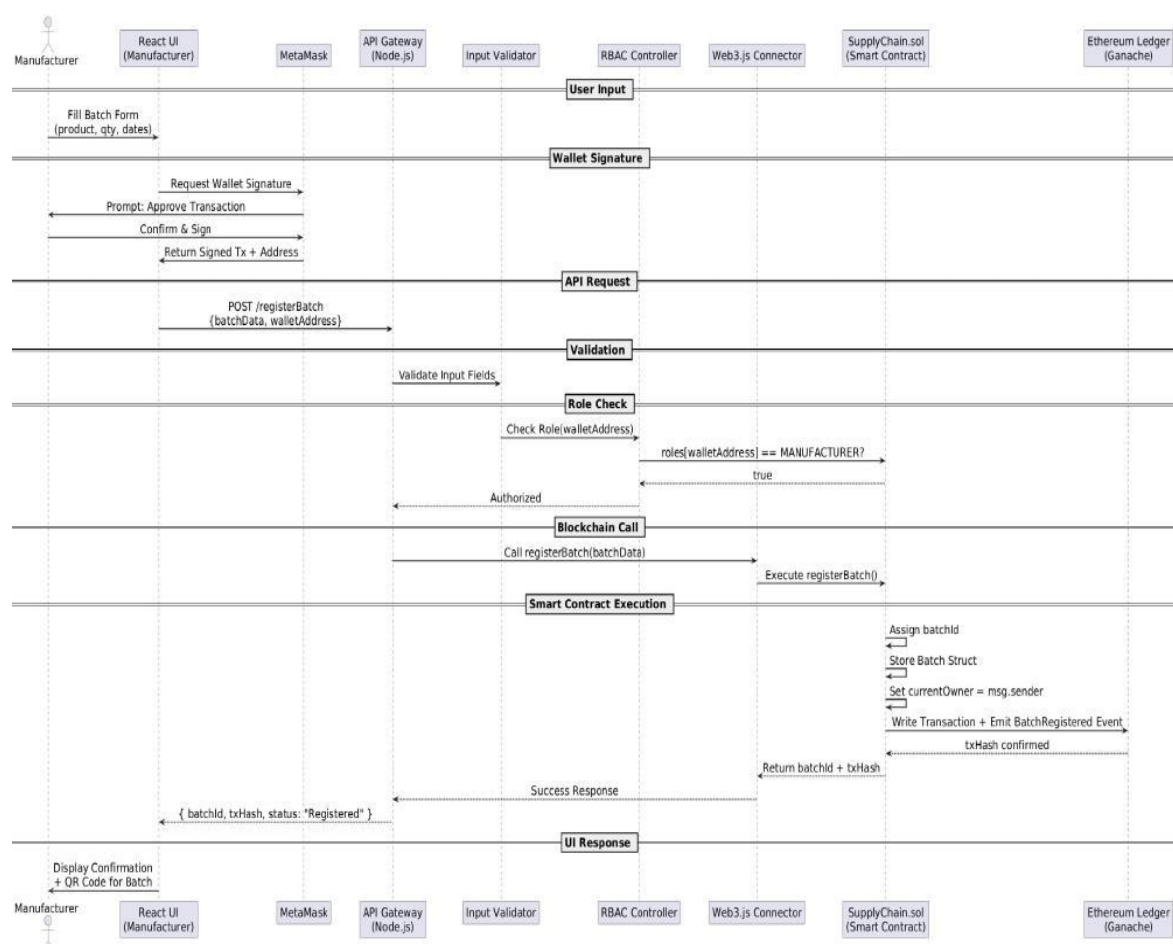


Figure 6.8 Sequence Diagram - Batch Creation

Ownership Transfer

The figure 6.9 illustrates how ownership of a medicine batch is securely transferred by the transporter in the blockchain-based system. The process begins when the transporter enters the batch ID and new owner details, then signs the transaction using MetaMask. The request is sent through the API, where role verification ensures that only an authorized transporter can perform the action. Once verified, the smart contract executes the ownership transfer, updates the blockchain record, and logs the transaction. Finally, the system confirms the transfer and updates the ownership chain, ensuring transparency, security, and tamper-proof tracking.

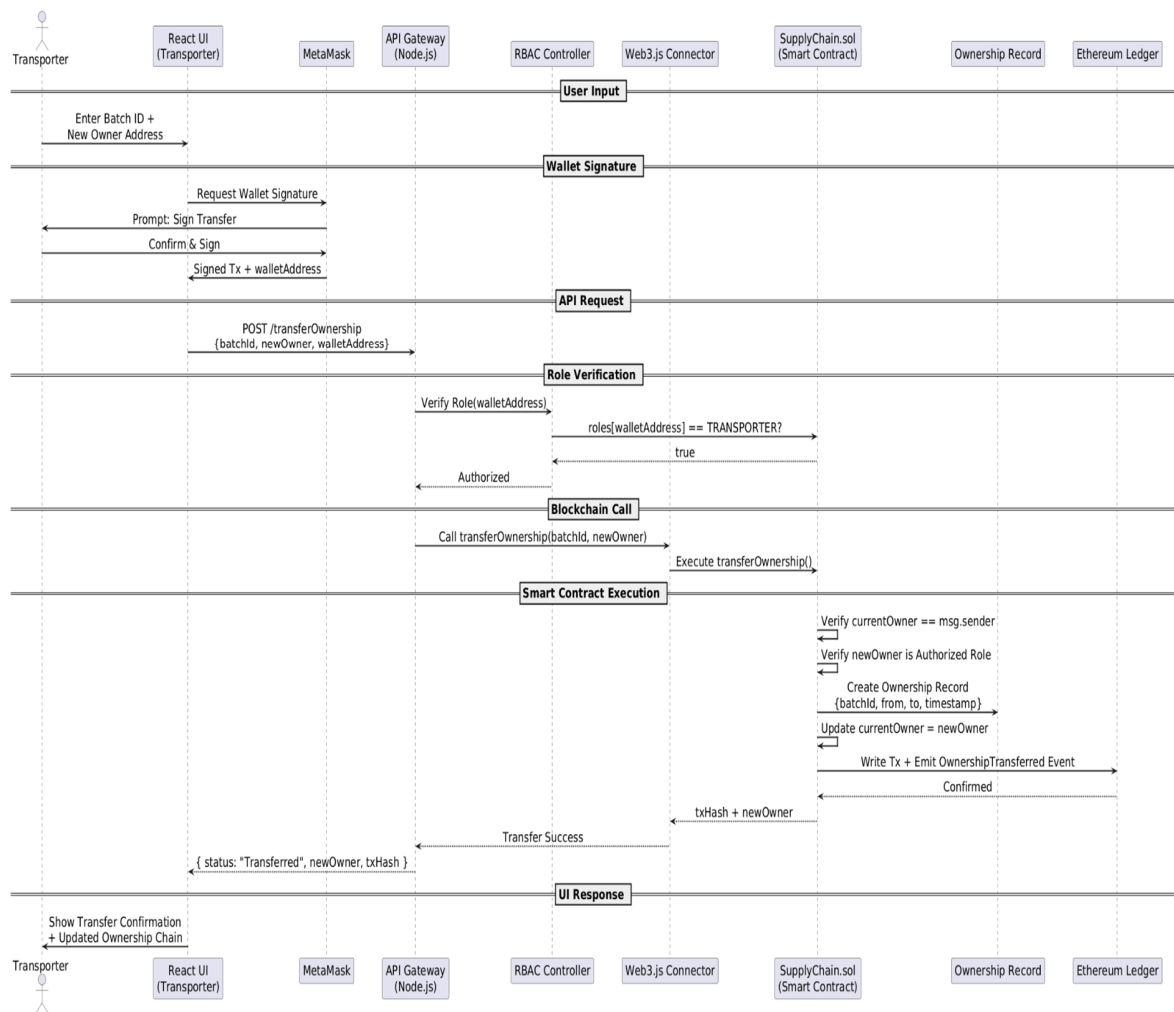


Figure 6.9 Sequence Diagram - Ownership Transfer

Product Verification

The figure 6.10 illustrates how a consumer verifies a product using the blockchain-based system. The process begins when the user scans a QR code through the React interface, which extracts the batch ID and sends it to the backend via an API. The request is then passed through Web3 to the smart contract, which retrieves product details from the Ethereum blockchain, checks authenticity, ownership history, and expiry. Based on the result, the system displays whether the product is authentic or not, ensuring transparency and preventing counterfeit medicines.

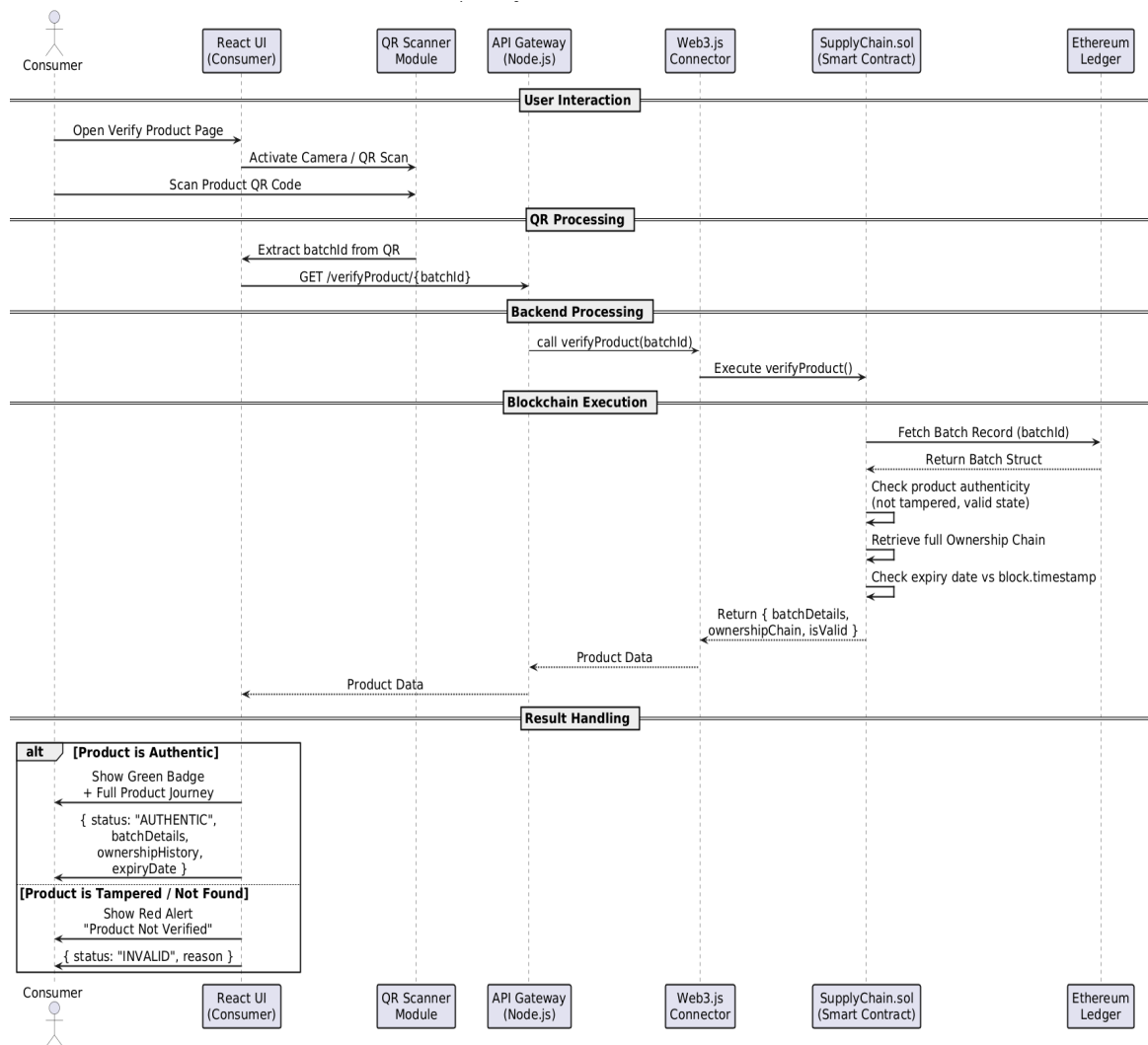


Figure 6.10 Sequence Diagram - Product Verification

Chapter 7

Applications

The proposed blockchain-based pharmaceutical supply chain management system has several practical applications in the healthcare and logistics sectors. It can be used to improve the safety, transparency, and efficiency of medicine distribution. Some of the key applications are as follows:

- **Counterfeit Drug Prevention:** By maintaining a tamper-proof record of all transactions, the system helps in identifying and preventing counterfeit medicines. Stakeholders and consumers can verify the authenticity of drugs before usage.
- **Pharmaceutical Supply Chain Management:** The system can be used by pharmaceutical companies to track medicines from manufacturing to final delivery. It ensures complete visibility of the product lifecycle and helps in efficient supply chain management.
- **Product Traceability and Verification:** The system allows users to trace the entire history of a medicine batch, including its origin, ownership transfers, and current status. This is useful for quality assurance and regulatory compliance.
- **Regulatory Compliance and Auditing:** Government authorities and regulatory bodies can use the system to monitor pharmaceutical activities and ensure that companies follow proper standards and guidelines. The transparent nature of blockchain simplifies auditing processes.
- **Efficient Recall Management:** In case of defective or expired medicines, the system enables quick identification of affected batches and their locations. This helps in faster and more efficient product recalls, reducing risks to public health.
- **Healthcare Safety Enhancement:** Hospitals and pharmacies can use the system to ensure that they are dispensing genuine and safe medicines, thereby improving patient safety and trust in healthcare services.

- **Integration with Future Technologies:** The system can be extended with technologies like IoT for real-time monitoring (e.g., temperature tracking of medicines) and QR codes for easy scanning and verification by end users.
- **Inventory and Stock Management:** The system can help pharmacies, distributors, and manufacturers maintain accurate inventory records and monitor medicine stock levels in real time. This reduces issues such as overstocking, stock shortages, and wastage of medicines.
- **Data Security and Secure Information Sharing:** Since blockchain provides a decentralized and encrypted environment, sensitive pharmaceutical data can be shared securely among authorized stakeholders. This improves collaboration between manufacturers, transporters, pharmacies, and healthcare organizations while maintaining data privacy and integrity.

Chapter 8

Conclusions and Future Enhancements

The proposed blockchain-based pharmaceutical supply chain management system provides a secure, transparent, and reliable solution to the major challenges faced in traditional pharmaceutical supply chains, including lack of transparency, inefficient tracking methods, and the increasing circulation of counterfeit medicines. By utilizing blockchain technology, the system ensures that all transactions are stored in a decentralized and tamper-proof environment, which guarantees data integrity and prevents unauthorized modifications. This creates trust and accountability among all stakeholders involved in the supply chain process.

The implementation of smart contracts automates important operations such as medicine batch creation, validation, and ownership transfer. Automation reduces manual intervention, minimizes human errors, and improves the efficiency of the overall system. The system also enables real-time product tracking and maintains a complete transaction history, thereby improving visibility and traceability throughout the supply chain lifecycle.

Another significant feature of the system is medicine authenticity verification. By allowing stakeholders and consumers to verify product details through blockchain records, the system helps in identifying genuine medicines and reducing the spread of counterfeit drugs. This directly contributes to improved patient safety and healthcare reliability. The integration of a user-friendly frontend interface along with MetaMask support further simplifies interaction with the blockchain network, making the system practical and accessible even for non-technical users.

The current implementation successfully demonstrates essential functionalities such as medicine batch registration, ownership transfer, blockchain storage, smart contract-based transaction management, and real-time verification. Although the system establishes a strong foundation, several future enhancements can further improve its scalability, usability, and real-world applicability.

In future versions, role-based stakeholder management can be implemented with separate dashboards for manufacturers, distributors, retailers, and administrators to improve system organization and security. Integration of QR code or RFID technology can make

medicine verification faster and more convenient for consumers and healthcare providers. IoT device integration can also be introduced to monitor environmental conditions such as temperature and humidity during transportation and storage, ensuring medicine quality and safety.

The system can further be deployed on public or private Ethereum networks for large-scale real-world implementation. Additional optimization and security techniques may also be applied to improve transaction speed, scalability, and performance. The frontend interface can be enhanced with a more responsive design and dedicated mobile application support for easier accessibility.

Future improvements may additionally include advanced analytics and reporting features, automated notifications and alert systems, integration with hospital and pharmacy management systems, cloud database support for storing supplementary non-sensitive information, and AI-based prediction and monitoring for supply chain analysis.

Overall, the proposed system demonstrates how blockchain technology can effectively transform pharmaceutical supply chain management by improving transparency, security, traceability, and efficiency. With further enhancements and large-scale deployment, the system has strong potential to become a complete real-world healthcare solution that ensures safe, authentic, and high-quality medicines reach the right people at the right time

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